Binomial Coefficients made easy

Binomial coefficients $\binom{n}{k}$ are the number of ways to select a set of k elements from n different elements without taking into account the order of arrangement of these elements (i.e., the number of unordered sets).

Binomial coefficients are also the coefficients in the expansion of $(a+b)^n$ (so-called binomial theorem):

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{k}a^{n-k}b^k + \cdots + \binom{n}{n}b^n$$

$$\begin{aligned}
(x+y)^0 &= 1\\
(x+y)^1 &= x+y\\
(x+y)^2 &= x^2+2xy+y^2\\
(x+y)^3 &= x^3+3x^2y+3xy^2+y^3\\
(x+y)^4 &= x^4+4x^3y+6x^2y^2+4xy^3+y^4\\
(x+y)^5 &= x^5+5x^4y+10x^3y^2+10x^2y^3+5xy^4+y^5
\end{aligned}$$

Analytic formula for the calculation:

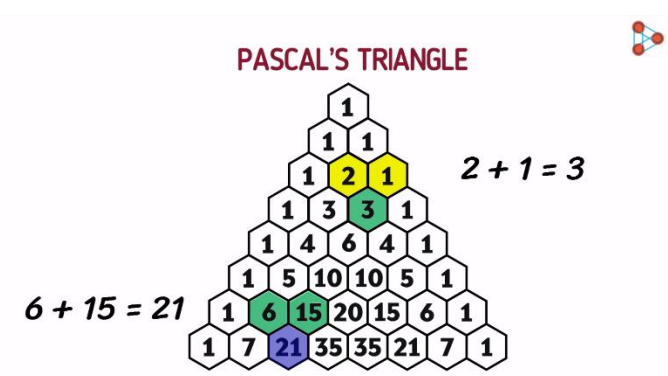
$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

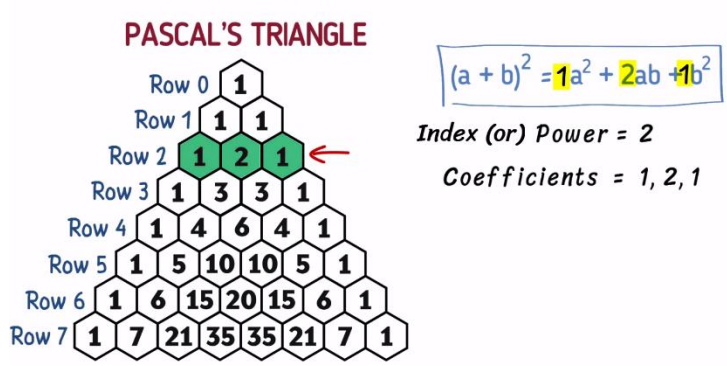
This formula can be easily deduced from the problem of ordered arrangement (number of ways to select k different elements from n different elements). First, let's count the number of ordered selections of k elements. There are n ways to select the first element, $n-1$ ways to select the second element, $n-2$ ways to select the third element, and so on. As a result, we get the formula of the number of ordered arrangements: $n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!}$. We can easily move to unordered arrangements, noting that each unordered arrangement corresponds to exactly $k!$ ordered arrangements ($k!$ is the number of possible permutations of k elements). We get the final formula by dividing $\frac{n!}{(n-k)!}$ by $k!$.

Recurrence formula (which is associated with the famous "Pascal's Triangle"):

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

PASCAL TRAINGLE (n=rowId, k=columnId)





Straight forward calculation from analytical formula

```
int C(int n, int k) {
    int res = 1;
    for (int i = n - k + 1; i <= n; ++i)
        res *= i;
    for (int i = 1; i <= k; ++i)
        res /= i;
}
```

Why are we looping from n-k+1 to n?

$$^6C_2 = \frac{6!}{2!\,(6-2)!} = \frac{6 \times 5 \times \cancel{4!}}{2! \times \cancel{4!}}$$

But this will result in integer overflow. What we can do is use the symmetric identity of Binomial coefficients and also tweak the body of the for loop to multiply and divide at the same time.

$$\binom{n}{k} = \binom{n}{n-k}$$

So if K is bigger than N-K, we can take K = N-K. Because of the symmetry identity.

Note that in the above implementation numerator and denominator have the same number of factors (k). Simple reasoning.

For the numerator, n! 6! = 6 . 5 . 4 . 3 . 2 . 1

From the 6 integers, we are taking out n-k integers from back.

Here we are taking out (6-2) = 4! = 4 . 3 . 2 . 1 from the numerator. Hence we are left with only K integers in both numerator and denominator. Here we are left with (6, 5) in numerator and (1, 2) in denominator.

So we can run a loop for k times.

```
int C(int n, int k) {
    if(k>n-k)
        k = n-k;
    int res = 1;
    for (int i = 1; i <= k; i++){
        int numerator = n-i+1;
        int denominator = i;
        res = (res * numerator)/denominator;
    }
}
```

Nth Catalan Number

$$\text{The } n\text{th Catalan number } C_n = \frac{1}{n+1}\binom{2n}{n}$$